

MANJARI ANANT

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EDUCATION

JOHNS HOPKINS UNIVERSITY

Baltimore, MD

Doctor of Philosophy, Neuroscience

August 2021 – (exp) May 2027

UNIVERSITY OF WASHINGTON, GPA: 3.9

Seattle, WA

Bachelor of Science, Bioengineering, Mathematics Minor

June 2021

RESEARCH EXPERIENCE

KEBSCHULL AND FAN LABS, JOHNS HOPKINS UNIVERSITY

Baltimore, MD

Graduate Student Researcher, Neuroscience

July 2022 – present

- Performing spatial transcriptomics (ST) experiments on brain tissue (mice, chickens) to study cerebellar nuclei development and evolution
- Co-developing “STalign” Python-based ST image 2D/3D alignment method
- Creating computational pipeline to visualize gene expression from ST imaging data
- Mentoring one master’s student and two undergraduate students

PUBLICATIONS

(* = First Author, # = Corresponding Author)

First author publications:

1. **Anant MMG**, Zuercher Z, Lowman M, Shi C, Faltine-Gonzalez D, Piacentino ML, Fan J, Kechschull JM. Excitatory progenitors drive cerebellar nuclei development and evolution. *bioRxiv*, 2026
2. Qi H*, **Anant MMG***, Faltine-Gonzalez D*, BARseq3: a modular system for integrating spatial omics and cellular barcoding in single cells. *bioRxiv*, 2026
3. Clifton K*, **Anant MMG*** et al. Alignment of spatial transcriptomics data using diffeomorphic metric mapping. *Nature Communications*, 2023
4. Koneru R*, **Anant MMG*#**, Kim H, Cheron J, Kechschull JM#. Alignment of spatial transcriptomics data using diffeomorphic metric mapping. *Journal of Open Source Software*, 2025
5. **Anant MMG***, Faltine-Gonzalez D*, Lavaca S*, Kechschull JM. Principles of brain region evolution: insights from the cerebellar nuclei. *Annual Review of Neuroscience*, 2025

Co-author publications:

6. Cheron J, Lowman M, **Anant MMG**, Siau M, Kechschull JM. Generation and validation of an Acan-Cre mouse line to selectively label Class-B excitatory neurons of the cerebellar nuclei. *The Cerebellum*, 2026
7. dos Santos Peixoto R, Miller B, Brusko M, Atta L, Aihara G, **Anant MMG**, ..., Fan J. Characterizing cell-type spatial relationships across length scales in spatially resolved omics data. *Nature Communications*, 2025
8. Atta L, Clifton K, **Anant MMG**, Aihara G, Fan J. Gene count normalization in single-cell imaging-based spatially resolved transcriptomics. *Genome Biology*, 2024
9. Stouffer KM, Trouve A, Younes L, Kunst M, Ng L, Zeng H, **Anant MMG**, Fan J, Kim Y, Miller MI. A Universal Method for Crossing Molecular and Atlas Modalities using

Simplex-Based Image Varifolds and Quadratic Programming. *Nature Communications*, 2023

10. Zhen L, Darrow R, Chen N, **Anant MMG**, Tang C, Gorantla L, Crawford L, Dryg I, Ratner B. Soft, precision-engineered porous, hydrogel scaffolds mechanically tailored towards applications in the central nervous system. *Journal of Bioactive and Compatible Polymers*, 2024

TALKS

1. Invited Seminar, Kalhor Lab Group, Johns Hopkins University, September 2026
2. Center for Computational Biology, Johns Hopkins University, March 2026
3. Center for Computational Biology, Johns Hopkins University, April 2025
4. Guest Lecture, EN.580.428: Genomic Data Visualization, Johns Hopkins University, March 2024
5. Single Cell Symposium, Johns Hopkins School of Medicine, November 2023
6. Center for Computational Biology, Johns Hopkins University, March 2023
7. Guest Lecture, EN.580.428: Genomic Data Visualization, Johns Hopkins University, March 2023

POSTER PRESENTATIONS

Graduate:

Transcriptomic Patterns during cerebellar nuclei development. (**Anant MMG**, Faltine-Gonzalez D, Tward D, Fan J, Kebschull J.)

1. *Gordon Research Conference Cerebellum*, July 2025
2. *Kavli Neuroscience Discovery Institute Symposium*, April 2025
3. *One Neuro Speaker Series: Justus Kebschull*, February 2025
4. *Cerebellar Development and Disease at Single-Cell Resolution Conference*, September 2024
5. *Gordon Research Conference Cerebellum*, August 2023

Undergraduate:

6. **Anant MMG**, Burch RM, Perlmutter SI, Fetz EE, Moorjani S. Theta-Burst Stimulation for Strengthening Cortical Connections. *Neural Computation and Engineering, Connection*, January 2020
7. **Anant MMG**, Ratner B. Optimizing Biocompatibility and Conductivity of Brain-Computer Interfaces. *University of Washington Research Symposium*, May 2019

AWARDS AND HONORS

1. 2024: Kavli Distinguished Pre-doctoral Fellowship
2. 2023: Gordon Research Conference Cerebellum Poster Prize
3. 2023: NSF GFRP Honorable Mention

OUTREACH

1. Kavli Multi-omics Seminar Organizer, Johns Hopkins University, 2024-2026
2. Graduate Teaching Assistant, Neuroscience and Cognition I, Johns Hopkins School of Medicine, 2022
3. Volunteer, Science in Action, Johns Hopkins School of Medicine, 2022 – 2023
4. Undergraduate Teaching Assistant, BIOEN 316: Signals and Sensors, University of Washington, 2020 – 2021
5. Vice President, Neural Engineering Club, University of Washington, 2018-2021